

ANDRÉS FELIPE GACHARNÁ TIBOCHA

Software Engineer | Backend & Distributed Systems

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PROFESSIONAL SUMMARY

Systems Engineer specializing in backend engineering, distributed systems, and scalable software architecture. Proven experience designing and implementing decoupled microservices using **NestJS**, **C# (.NET Core)**, **TypeScript**, and **Python** for enterprise retail, healthcare, and logistics operations. Solid background in **Clean Architecture**, relational and NoSQL database modeling (**PostgreSQL**, **SQL Server**, **MongoDB**), and containerized Linux environments, with complementary frontend experience in **React**.

TECHNICAL SKILLS

Languages & Core Frameworks: TypeScript, C# (.NET / ASP.NET Core), Node.js, NestJS, Python, React.

Architecture & Backend: Microservices, Clean Architecture, RESTful APIs, GraphQL, Entity Framework, SOLID Principles, Design Patterns, OpenAPI (Swagger), LLM & API Integration.

Databases & Data Management: PostgreSQL, SQL Server, MongoDB, Firebase.

Infrastructure, Cloud & DevOps: Linux (Bash scripting), Docker, CI/CD Pipelines, Proxmox VE (Virtualization & LXC), Nginx, Cloudflare Tunnels, GCP, Kubernetes.

WORK EXPERIENCE

GEEKCORE S.A.S.

December 2024 – Present

Software Engineer

Bogotá, Colombia

- **Architected and deployed high-throughput GraphQL and RESTful APIs** using NestJS and ASP.NET Core, automating personnel management workflows and centralizing supplier/store data to cut reporting turnaround by ~70% for enterprise retail operations (Decathlon).
- **Decomposed a legacy enterprise monolith into a modular microservices architecture**, driving horizontal scalability and enabling platform rollout across 3 international markets beyond initial projections.
- **Engineered mission-critical tracking services for healthcare logistics**, designing optimized PostgreSQL relational schemas and automated CI/CD pipelines that delivered continuous availability and real-time operational visibility.
- **Implemented automated unit and integration testing suites** adhering strictly to Clean Architecture and SOLID principles, improving release stability and code maintainability across multi-tenant production systems.
- **Drove technical delivery within cross-functional Agile workflows**, actively participating in sprint planning, architecture reviews, and backlog refinements to ensure reliable delivery cycles.

PROJECTS & HONORS

- **Cloud & Self-Hosted Infrastructure (February 2026)**
 - **Architected a high-availability virtualization environment** utilizing Proxmox VE, Debian LXC containers, and Docker to host and test containerized microservices and backend applications.
 - **Configured secure zero-trust networking** implementing Cloudflare Tunnels, custom DNS, automated SSL/TLS termination via Nginx Proxy Manager, and internal subnet isolation for service decoupling.
 - **Automated self-hosted deployments** for developer tools and production-like staging databases (PostgreSQL, Redis), ensuring environment parity and efficient resource utilization.
- **1st Place Winner — RISKTECH S.A.S. Corporate Hackathon (November 2025)**
 - **Architected an end-to-end real-time fraud detection engine** for banking POS transactions, building a high-throughput ingestion API with NestJS, Python, and PostgreSQL.
 - **Engineered a tiered low-latency ML pipeline:** deployed a fast CatBoost classification model for sub-second scoring, with automated fallback routing to a BERT encoder for deep contextual arbitration on borderline fraud cases.
 - **Implemented automated data enrichment workflows** utilizing LLM APIs to synthesize complex transactional features into structured JSON for downstream analytics under strict latency constraints.
 - **Secured 1st place** among participating university research teams by delivering a functional, resilient architecture under a high-pressure 48-hour deadline.

EDUCATION

Universidad Jorge Tadeo Lozano

Bachelor of Science in Systems Engineering

Bogotá, Colombia

Graduated March 2026

CERTIFICATIONS

- **Fortinet Certified Associate in Cybersecurity** — Fortinet (Nov 2025)
- **Scrum Fundamentals Certified** — SCRUMstudy (Nov 2025)
- **DevOps Foundations: Core Principles and Practices** — Microsoft (Jul 2025)

LANGUAGES

Spanish: Native **English:** Professional Working Proficiency (B2)